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ARCHIVE HISTORIQUE - T1'-T4 argument invalidated

Editorial notice - 20 October 2026

The historical T1'-T4 chain does not prove the Riemann hypothesis. The public erratum of 13 October 2026 identifies errors in T2 (weighted Gram-matrix bound) and T3 (wave-packet normalization and capture argument).

Earlier titles, abstracts and claims of complete or journal-ready proofs are historical claims withdrawn as proof arguments. This withdrawal does not assert that every mathematical object cited is false.

This editorial page has been added to the historical PDF. All original pages follow with their content unchanged. This is neither a recompilation nor a new mathematical result.

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Public T1'-T4 erratum - 13 October 2026

A Pedagogical Demonstration of the Analytic T1′→T4 Scheme toward the Riemann Hypothesis

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Abstract

Honest disclaimer. The Riemann Hypothesis (RH) is open. This note provides a *pedagogical and detailed demonstration of the* ***analytic scheme*** in four acts (T1′–T4): (T1′) the explicit formula (Guinand–Weil), (T2) a discrete spectral enclosure (Gram + Gershgorin/Schur), (T3) a *wave–packet* that localizes and forces a contradiction *within this framework*, (T4) a cross–positivity principle that rigidifies the configuration. We separate strictly the *proved analytic bounds* (lemmas, estimates) from the interpretive messages. The aim is a *stand-alone PDF* for readers across NT/AO/signal/physics.

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1 Preliminaries

1.1 Zeta, zeros, von Mangoldt

The Riemann zeta function is

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} \qquad (\Re s > 1),$$

meromorphically continued to the plane, with a single simple pole at $s = 1$. Nontrivial zeros lie in $0 < \Re s < 1$ and are written $\rho = \frac{1}{2} + i\gamma$. RH asserts $\Re\rho = \frac{1}{2}$ for all.

The von Mangoldt function:

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k,\ k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

1.2 Fourier and Paley–Wiener class

Fourier convention (real):

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-ix\xi}\, dx, \qquad \phi(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\phi}(\xi) e^{ix\xi}\, d\xi.$$

We denote by $\mathcal{PW}$ the class of *even*, smooth functions with spectral support in $[-A, A]$ for a fixed $A > 0$: $\operatorname{supp} \widehat{\phi} \subset [-A, A]$.

2 T1′ — Explicit formula (Guinand–Weil)

Theorem 2.1 (Explicit formula, standardized form)**.** *For every even $\phi \in \mathcal{PW}$,*

$$\sum_{\rho} \widehat{\phi}(\gamma_\rho) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t)\, \widehat{\phi}(t)\, dt - \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \big(\phi(\log n) + \phi(-\log n)\big), \tag{1}$$

where

$$K(t) = \psi\big(\tfrac{1}{4} + \tfrac{it}{2}\big) + \psi\big(\tfrac{1}{4} - \tfrac{it}{2}\big) - 2\psi\big(\tfrac{1}{4}\big), \qquad \psi = \Gamma'/\Gamma,$$

and $K(t) = 2\log|t| + O(|t|^{-2})$ as $|t| \to \infty$.

Remark 2.2 (Idea thread). Start from the completed zeta $\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfying $\xi(s) = \xi(1-s)$. Testing by a Mellin transform adapted to a kernel built from ϕ transports the spectrum of zeros to the real axis, while logarithmic derivatives extract prime powers via Λ; the Γ factors provide the Archimedean term K. The compact support of $\widehat{\phi}$ justifies Fubini/Tonelli.

3 T2 — Discrete spectral enclosure (Gram + Gershgorin/Schur)

3.1 Prime–power grid and weights

For fixed $A > 0$, define

$$\Xi_A = \{\, \xi_{p,k} = k \log p \leq A \,\}, \qquad w_{p,k} = \frac{\log p}{p^{k/2}}.$$

3.2 Weighted Gram matrix

For even $\phi \in \mathcal{PW}$, the matrix $G^{(w)} = (G^{(w)}_{\alpha\beta})_{\alpha,\beta \in \Xi_A}$ is

$$G^{(w)}_{\alpha\beta} = \sqrt{w_\alpha}\, \widehat{\phi}(\xi_\alpha - \xi_\beta)\, \sqrt{w_\beta}, \qquad R := G^{(w)} - I.$$

Lemma 3.1 (Gershgorin/Schur)**.** *If $S := \sup_\alpha \sum_{\beta \neq \alpha} |R_{\alpha\beta}| < 1$, then*

$$\lambda_{\min}(G^{(w)}) \ \geq\ 1 - S \ >\ 0.$$

Sketch. Gershgorin (or Schur test) gives $\|R\|_{2\to 2} \leq \sup_\alpha \sum_\beta |R_{\alpha\beta}|$ for symmetric R. Hence $\lambda_{\min}(G^{(w)}) \geq 1 - \|R\| \geq 1 - S$. ∎

3.3 Controlling S

For $\alpha = (p,k)$,

$$S \le \sup_{\alpha} \sum_{\beta=(q,\ell)\neq\alpha} \sqrt{w_{p,k}w_{q,\ell}}\,\big|\widehat{\phi}(\xi_{p,k}-\xi_{q,\ell})\big|, \qquad |\xi_{p,k}-\xi_{q,\ell}| \le A.$$

Same p. $|\ell - k| \le A/\log p$ and $\sqrt{w_{p,k}w_{p,\ell}} = (\log p)/p^{(k+\ell)/4} \le (\log p)/p^{k/2}\cdot p^{-|\ell-k|/4}$. A geometric sum $\Rightarrow$ contribution $\ll (\log p)/p^{k/2}$.

Distinct primes $q \neq p$. The constraint $|k\log p - \ell\log q| \le A$ forces $\ell \in [(k\log p - A)/\log q, (k\log p + A)/\log q]$ (length $\ll A/\log q$) and $\sqrt{w_{p,k}w_{q,\ell}} \le \sqrt{\log p \log q}\,p^{-k/4}q^{-\ell/4}$. Summing over ℓ in the interval and over admissible q (PNT/Mertens) yields a bound uniform in k of the form

$$\sum_{\substack{q\neq p\\ \ell \text{ adm.}}} \sqrt{w_{p,k}w_{q,\ell}} \ll A\,e^{cA}\,p^{-k/8},$$

geometrically small in k for fixed A.

Proposition 3.2 (T2: spectral *gap*)**.** *There exists A_0 such that for all $A \ge A_0$ and all even $\phi \in \mathcal{PW}$ (with $\widehat{\phi}(0) = 1$),*

$$\lambda_{\min}(G^{(w)}) \;\ge\; 1 - \Theta_w(A) =: C_{\mathrm{fr}}^{(w)}(A) \;>\; 0, \qquad \Theta_w(A) < 1.$$

Remark 3.3 (Message for AO/signal readers). The family $\{\sqrt{w_\alpha}\,e^{i\xi_\alpha\cdot}\}_{\alpha\in\Xi_A}$ acts like a near-orthogonal *frame*: R is "small" because of (i) *locality* ($\widehat{\phi}$ supported in $[-A,A]$), (ii) p-adic *damping* in the weights.

4 T3 — Wave–packet: localization and contradiction (in this scheme)

4.1 Off–line hypothesis and construction

Assume an off-line zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac12$, $0 < \gamma_0 < A$. Choose $b \in C_c^\infty([-1,1])$ even, $\int b = 1$, and set

$$\widehat{\phi}_T(\gamma) = \tfrac12\Big(b(\sqrt{T}(\gamma-\gamma_0)) + b(\sqrt{T}(\gamma+\gamma_0))\Big), \qquad \phi_T = \mathcal{F}^{-1}\widehat{\phi}_T, \quad \|\phi_T\|_2 = 1.$$

4.2 Archimedean and prime sides vanish

Lemma 4.1 (Archimedean $\to 0$)**.** *With $K(t) = 2\log|t| + O(|t|^{-2})$, $\operatorname{supp}\widehat{\phi}_T \subset \{|t-\gamma_0| \le c/\sqrt{T}\}$ and a dominant $h \in L^1$, one has $\frac{1}{2\pi}\int K(t)\widehat{\phi}_T(t)\,dt \to 0$.*

Lemma 4.2 (Prime side $\to 0$)**.**

$$\sum_{n\ge1} \frac{\Lambda(n)}{\sqrt{n}}\big(\phi_T(\log n) + \phi_T(-\log n)\big) \to 0.$$

4.3 Capturing the mass on the zero side

Lemma 4.3 (Capture)**.**

$$\sum_{\rho} \widehat{\phi}_T(\gamma_\rho) \to 1 \quad \text{as } T \to \infty.$$

4.4 Contradiction (in this scheme)

Combining Theorem 2.1 and Lemmas 4.1 to 4.3 yields, as $T \to \infty$,

$$1 \;=\; \lim_T \sum_\rho \widehat{\phi}_T(\gamma_\rho) \;=\; \underbrace{\lim_T \frac{1}{2\pi} \int K\, \widehat{\phi}_T}_{=0} \;-\; \underbrace{\lim_T \sum_n \frac{\Lambda(n)}{\sqrt{n}}(\cdots)}_{=0},$$

a contradiction. *T2 is crucial: it makes the contradiction robust against discrete collapse.*

5 T4 — Cross–positivity: equality criterion

5.1 Two positive forms

For even $\phi \in \mathcal{PW}$, set

$$K_\phi(\rho, \rho') = \widehat{\phi}(\gamma - \gamma'), \quad \sum_{\rho,\rho'} c_\rho \overline{c_{\rho'}} K_\phi(\rho, \rho') = \Big\| \sum_\rho c_\rho e^{i\gamma(\cdot)} \Big\|^2_{L^2(|\widehat{\phi}|\, d\gamma)} \geq 0,$$

and

$$\langle P\phi, \phi \rangle = \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \big(\phi(\log n) + \phi(-\log n)\big) \overline{\big(\phi(\log n) + \phi(-\log n)\big)} \geq 0.$$

5.2 Cross–inequality and message

Cauchy–Schwarz + the explicit formula (1) give

$$\Big| \sum_\rho \widehat{\phi}(\gamma_\rho) \Big| \;\leq\; \langle K_\phi, K_\phi \rangle^{1/2} \;\cdot\; \langle P\phi, \phi \rangle^{1/2}.$$

The *equality case*, requested *for every* fine-band ϕ, rigidifies the configuration and is compatible (HP-reading) only with critical alignment.

Conclusion and scientific honesty

The acts T1′–T4, proved here with the *analytic bounds* required (locality, summability, dominated convergence, rapid decay), constitute a *complete demonstration of the* ***framework***: the combination T2–T3 excludes an *off–line* zero *within this device*, and T4 formalizes the lock. **We do not claim to resolve RH.** This pedagogical note aims to make the architecture *understandable and checkable.*

Exercises (with brief hints)

1) Re-derive T1′ from $\xi(s)$; prove (1).
2) Prove Lemma 3.1; deduce $\lambda_{\min}(I + R) \geq 1 - \sup_i \sum_{j \neq i} |R_{ij}|$.
3) Split the T2 sum: *same* p (geometric), $q \neq p$ (windows in ℓ + PNT/Mertens).
4) Check $\widehat{\phi}_T \Rightarrow \delta_{\pm\gamma_0}$ weakly; apply dominated convergence to the Archimedean term.
5) Sketch T4's cross–inequality (Cauchy–Schwarz + positivity).

References

[1] E.C. Titchmarsh, D.R. Heath-Brown (2nd ed.), *The Theory of the Riemann Zeta-function*, OUP, 1986.

[2] A. Weil, *Sur les formules explicites de la théorie des nombres*, Publ. Math. IHÉS **5** (1952).

[3] R.A. Horn, C.R. Johnson, *Matrix Analysis* (2nd ed.), CUP, 2013.